Designing Accessible Technology with Adults with Down Syndrome

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General Technology Usage

RQ How do adults with Down syndrome use technology in daily life, and what opportunities exist for accessible design?

METHOD 6 adults with DS (ages 21–37) · 7 caregivers · 3 experts
Exploratory qualitative study · 1 hr in-person session with participants with Down syndrome + 30 min with caregivers + 1 hr with experts · Semi-structured interviews · Reflexive thematic analysis

KEY FINDINGS

Multi-device use

Participants used multiple devices across media and communication, with distinct roles for each device in their daily routines.

Technology not used at work

Assistive technology was absent from participants' workplaces; technology use was concentrated in home and social contexts.

Context-dependent communication

Participants adjusted their communication strategies to their social context (home, work, friends) rather than relying on speech assistive technology.

Family involvement across the lifecycle

Parents supported learning phases; participants valued independence in ongoing use and discovery, expressing a preference for *personal privacy* within family units.

Defining their own proficiency

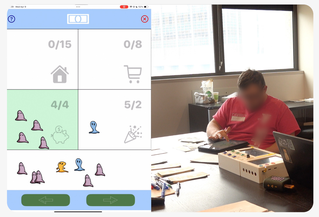
Participants' expectations for skill mastery differed from those of parents and experts; they wanted their existing skills recognized on their own terms.

Money Management

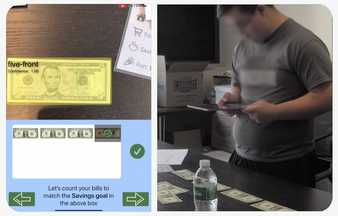
RQ How do different budgeting technology modalities align with adults with Down syndrome's prior experiences and error correction strategies?

METHOD 7 adults with DS (ages 19–39) · 5 support persons
Technology probe study · 2.5 hr in-person session · 3 probes in self-selected order · UES-SF + interview after each · Qualitative content analysis

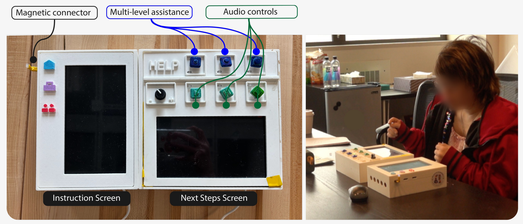
KEY FINDINGS



Gamification captured attention but shifted focus toward game progression over understanding.



Augmented Reality automation reduced arithmetic demands but also reduced error-checking.



Tangible device controls supported engagement and on-demand audio guidance, though dual-screen coordination and multi-path flows posed challenges.

Prior tech experience shaped interaction high-tech/low-budget users rushed leading to amore errors; low-tech users moved cautiously; strong budgeters resisted over-scaffolding.

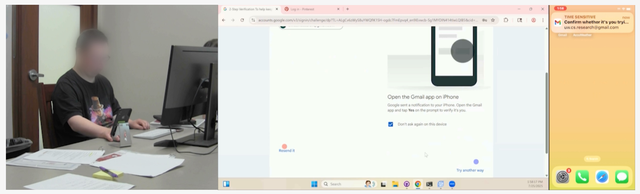
Error recovery relied on brief, situated human prompts linking screen state to physical cash. Interface feedback alone rarely prompted independent checking.

Accessible Security

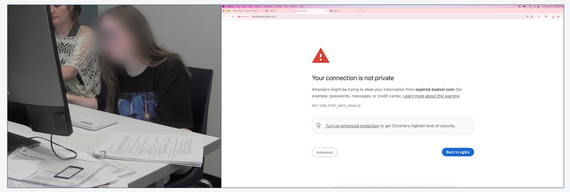
RQ How do adults with IDD encounter, interpret, and respond to security and privacy notifications?

METHOD 6 adults with IDD (5 DS, 1 ASD; ages 19–38) · 4 support persons
Usability study · 1–1.25 hr in-person session · 3 tasks with embedded notifications + semi-structured interviews · Eye tracking + click logs + audio/video · Reflexive thematic analysis

KEY FINDINGS



P2 surprised and confused by the 2FA notification



P5 looking for safe option for invalid HTTPS notification

Notification meaning was shaped by task context

Participants interpreted notifications through the lens of their current task.

Familiar concepts anchored understanding

Everyday ideas (e.g., cookies = food, tracking = following) served as primary frames for interpreting risk.

Uncertainty drove exploration or avoidance

When outcomes were unclear, participants either clicked exploratorily to test affordances or avoided interaction entirely. Both were protective strategies, but each carried unintended risk.

Overarching Design Considerations

Five principles for designing inclusive technology with and for adults with Down syndrome

01

Tangible & Multimodal

Physical controls, audio cues, and concrete feedback anchor abstract digital tasks, supporting engagement and reducing reliance on screen-only comprehension.

02

Age-appropriate, visual-first design

Concrete visual metaphors, familiar anchors, and iconicity can reduce cognitive load, but should not be infantilizing. Abstract or jargon-heavy language can fail even when simplified.

03

Action-oriented, outcome-previewing feedback

Users need to see not just what to do, but what will happen next. Predictive cues reduce avoidance and support safer decision-making.

04

Graduated scaffolding with context-aware design

Support should adapt to individual ability, experience, and confidence: neither over-scaffolding (collapsing autonomy) nor under-scaffolding (leaving users without guidance).

05

Interdependence as a first-class design choice

Design should support collaborative decision-making with trusted others while preserving agency, privacy, and autonomy for users with IDD.

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